

Which Signals Actually Support the Choice of a VAE Latent Dimension? A Pre-Registered Comparison of Intrinsic-Dimension, Activity-Based and Pruning Criteria

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Abstract

Choosing the latent dimension of a variational autoencoder is usually done by grid search. Cheaper criteria have been proposed, based on an intrinsic-dimension estimate of the data or on how many latent variables stay active after training. Under one protocol fixed in advance, we asked which of these signals actually supports the decision. We compared intrinsic-dimension guidance, two published selection algorithms, a pruning method with a reconstruction constraint, an automatic-relevance variant, grid search, coarse search and fixed dimensions, across four datasets, five seeds and five regularisation settings. Guidance by intrinsic dimension gave no measurable advantage: a plain ascending scan under the same quality criterion matched or beat it at lower cost on every dataset, and the guidance step declined to apply on two of three image datasets. The activity readout carried no information beyond a matched set of random features. Nearly every criterion was governed by a free threshold, and the dimension selected moves several-fold when that threshold varies; the ordering of methods by this sensitivity is itself reproducible, and one method is insensitive to it. We report these negative results, the sensitivity ordering, and the implementation failure modes we met, as the contribution.

Keywords: variational autoencoder; latent dimension; intrinsic dimension; active units; model selection; pruning; automatic relevance determination; reproducibility; negative results; pre-registration

1. Introduction

The latent dimension of a variational autoencoder [1] has to be set before training, and it is usually set by grid search: train a model at each candidate capacity and keep the one that performs best. This is reliable and expensive, and a number of cheaper criteria have been proposed in its place. Some estimate the intrinsic dimension of the data [9,10] and use it to place a search window [18]. Others train a single large model briefly and read off how many latent coordinates remain in use, either from the variance of the posterior means [25] or from a classification of the latent variables into types [19]. Others again prune during training, under an explicit constraint on reconstruction quality [20] or under a prior that suppresses irrelevant axes [23].

Each of these criteria has been reported to work on the datasets its authors used. What has not been established is how they behave against one another under a single protocol: with the same quality target, the same accounting of cost, the same number of seeds, and

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the same treatment of the cases where a method declines to answer. Without that, it is difficult to know whether a reported saving reflects the criterion or the conditions in which it was measured.

This paper reports such a comparison. We fixed the protocol in advance — the quality target, the budget, the termination states, the decision rules and the number of seeds — and recorded every subsequent change with its reason. We then ran intrinsic-dimension guidance, two published selection algorithms, a constrained pruning method, an automatic-relevance variant, grid search, coarse search and fixed dimensions across four datasets, five seeds and five settings of the regularisation weight.

Our results are largely negative, and we report them as the contribution. Guidance by an intrinsic-dimension estimate bought nothing that a plain ascending search under the same quality target did not, and it declined to apply on the datasets it was not developed on. The activity-based readout was no more stable than the readout it was meant to improve on, and in a prediction task fixed in advance it was not distinguishable from random features of the same number. The one criterion that transferred well is not ours: it coincides with a constraint already used by an existing pruning method, and we say so.

What does emerge is a measurement. Nearly every rule we examined is governed by a free parameter, and the dimension it selects moves severalfold when that parameter is varied. The methods differ in this sensitivity by two orders of magnitude, the ordering reproduces across datasets, and one method sits at zero. We also document three implementation defects of a single kind — a sum written as a mean — that left training intact and the resulting curves plausible. Section 6 presents the measurements and Section 7 discusses what would change the conclusion.

2. Related Work

Two overlaps with existing work bear on how this paper should be read, and we state them before the survey rather than after it.

First, the appendix of the intrinsic-dimension selection algorithm that we use as a comparison method [18] already proposes choosing the latent dimension from the variable types of a briefly trained model. That is the same idea as the activity-based verification this line of work set out to study. Its authors describe it and set it aside, reporting that variable types are unreliable early in training. Our measurements reproduce that instability, and extend the comparison to long training, where we find that the advantage they were weighing does not appear in either direction.

Second, the pruning method we added during revision [20] selects the smallest capacity that satisfies an explicit upper bound on reconstruction error. The quality criterion used throughout this paper is that construction. We do not present it as a contribution, and in Section 6.3 we report the correspondence quantitatively.

2.1. Dimensionality Reduction and Intrinsic Dimension

Reducing dimension by finding coordinates that preserve structure is an old problem. Principal component analysis [7] gives a linear answer; Isomap [5] and locally linear embedding [6] give nonlinear ones by preserving geodesic or neighbourhood structure. These methods produce an embedding, not a criterion for how many coordinates a separately trained generative model should have, which is the question here.

Estimating the intrinsic dimension of a sample is a related but distinct task. We use the two-nearest-neighbour estimator [9] as the primary estimator and a maximum-likelihood estimator [10] as a cross-check; other families exist [11–14], including graph-based estimators built from geodesic entropic graphs [15]. We did not implement a graph-based estimator: the measurements in Section 6.2 show that the disagreement we care

about is already visible between the two estimators we do use, and adding a third would not resolve it. Deep representations have been reported to have lower intrinsic dimension than their inputs [16,17], which is part of the motivation for reading a latent dimension off such an estimate.

2.2. Choosing the Latent Dimension

Two published algorithms select the dimension from an intrinsic-dimension estimate of the representations. The first compares the estimated dimension of the sampled and the mean representation of a briefly trained model and returns the largest capacity at which they have not yet diverged. The second, given in the appendix of the same work, starts from twice the data estimate, doubles capacity until latent variables of the unused type appear, and returns the count of the remaining ones. As noted above, the second is the construction this line of work set out to study independently.

2.3. Pruning During Training

A different strategy trains one large model and removes capacity while training. One approach [20] places a binary gate on each latent coordinate, penalises the number of open gates [22], and holds reconstruction error below a stated bound by a Lagrange multiplier updated from the constraint violation [21]; the number of gates left open is the answer. Because the bound is stated in data space rather than in the latent space, its authors argue it is easier to set than a regularisation weight — an argument our measurements support, and which is why we adopt the same construction as our common quality target. Another approach [23] replaces the fixed prior with a hierarchical one whose per-axis variance is fitted to the encoded data, and reads relevance from that variance; it introduces no threshold of the kind the other methods rely on, and Section 6.3 shows what that buys.

2.4. Active Units and Posterior Collapse

Counting the latent coordinates whose posterior means retain variance is a standard diagnostic for posterior collapse [24,25], and the classification of latent variables into active, mixed and passive types [19] is a refinement of it. Both are read from a trained model rather than used to choose its capacity; the step from diagnostic to selection criterion is the one this paper evaluates.

2.5. Relation to Our Earlier Conference Paper

A conference paper by the same authors [35] reported that the intrinsic dimension of the data is informative for bottleneck design and that latent estimates fall below input estimates, using fully connected models on a single dataset. It did not consider active units, did not compare against other selection methods, and did not vary the regularisation weight. The present paper shares the starting observation and differs in what it concludes: the comparison it adds does not support using that observation to place a search window.

3. Evaluation Framework

Every method is evaluated under the same target, the same accounting and the same set of outcomes. This section states them; the settings themselves were fixed before any of the experiments reported here were run, and each later change is listed with its date and reason in the supplementary material.

3.1. Quality Target

A candidate is acceptable if its validation reconstruction error does not exceed that of a pre-declared reference model by more than a stated tolerance. The reference is the largest capacity on the candidate grid, trained with the same model family, likelihood and

regularisation weight as the candidates, and its cost is charged. The tolerance is the larger of a relative margin, three times the seed-to-seed standard deviation at the reference, and an absolute floor that keeps the criterion meaningful when the reference error is near zero. The resulting target is fixed before any intervention and is not recomputed afterwards.

As noted in Section 2, this is the same construction as the reconstruction constraint of an existing pruning method, and we make no novelty claim for it. We state it explicitly because it is the common yardstick that makes the comparison possible: without a shared target, methods that optimise different objectives cannot be said to reach the same quality.

3.2. Cost Accounting

Every method is charged up to the point where a model is selected and trained ready for final evaluation. That includes reference-model training, intrinsic-dimension estimation, candidate training, the reference model required by the quality target, any re-search, probe fitting where a method uses labels, and evaluation. Methods whose search phase is deliberately short keep that advantage, because the final training at the selected dimension is charged to every method alike. Repeated requests for the same configuration within one run of one method are charged once.

3.3. Outcomes

A run ends in one of five states: it selected a capacity smaller than the reference; it found nothing smaller than the reference that met the target; its estimation step failed a reliability check and it fell back to a pre-declared conventional search; it exhausted its budget; or it returned a capacity that cannot be trained. Runs that exhaust the budget or return a degenerate capacity are reported in the breakdown and are not removed from the averages. We state this because a comparison that drops them silently will favour whichever method fails least gracefully.

4. Theoretical Background

Two known results motivate the quantities we measure. Neither is a contribution of this paper and neither is used as a guarantee.

A continuous, self-intersection-free reconstruction of data supported on a d -dimensional manifold requires a latent space at least as large as the manifold's embedding dimension, which is bounded above by $2d$ for smooth manifolds [29]. This gives a geometric reason to look near twice an estimate of the intrinsic dimension when placing a search window. It does not tell us that an estimate obtained from a particular representation corresponds to the manifold dimension, nor does it place a lower bound on the average reconstruction error of a finite sample at a given tolerance. We therefore treat the factor of two as a default under test and measure its sensitivity rather than deriving it.

In a linear-Gaussian surrogate with fixed observation noise, the latent coordinates that remain informative are those whose signal exceeds a threshold set by the regularisation weight [30]; related behaviour has been reported for trained models [28]. This gives a reason to expect that the count of active coordinates responds to that weight. It does not license reading the count as an estimate of a geometric dimension, and it does not transfer to the nonlinear models we train. Claims in the earlier version of this work that went beyond this — a Jacobian-rank argument, an unproved upper bound, an effective-noise quantity defined in terms of the behaviour it was meant to explain, an identification of the activity count with the variable-type classification, and an inference of direction from the sign of a divergence term — have been removed.

5. Experimental Setup

5.1. Objective and Likelihood

The reconstruction term is summed over input dimensions and averaged over the batch; the divergence term is summed over latent dimensions and averaged over the batch [2]. This matters: an earlier version of this work averaged the divergence over latent dimensions as well, which made the effective regularisation weight inversely proportional to the very capacity being varied. Every variational experiment in that version was affected, and all of them have been re-run. A regression test now checks that the divergence scales with capacity for fixed per-coordinate statistics.

The likelihood follows the data type: Bernoulli for the binary image dataset [33], Gaussian elsewhere. This is stated in every table and figure, loss values are never ranked across datasets with different likelihoods, and the same weight under different likelihoods is not described as the same strength. The per-pixel mean squared error used for the quality target is common to all of them.

5.2. Data, Splits and Repetition

The datasets are MNIST [31], Fashion-MNIST [32], dSprites [33] and CIFAR-10 [34]. Official training data is split into training and validation portions by a fixed random draw whose indices are stored; the official test split is used once, on the checkpoint that validation selected. Subsets are drawn at random rather than taken from the head of the file. Transformations are fitted on the training portion only. The main comparison uses five independent seeds, and the whole procedure — reference models, estimation, selection, final training — is repeated for each; evaluating one checkpoint five times is not five seeds.

5.3. Models, Grids and Comparison Methods

Fully connected encoders and decoders are used for the two grayscale digit-style datasets and convolutional ones for the other two, with a candidate grid of twelve capacities for the first three datasets and ten for the largest. Table 1 defines the comparison methods.

Table 1. Comparison methods. Every method is charged for a final training at the capacity it selects, so that a short search phase is reflected in the cost rather than removed from it.

Method	Selection rule
Grid search	Train every capacity on the grid; keep the best validation reconstruction. Variants use the evidence lower bound or a linear probe.
Coarse search	Four log-spaced capacities from the same grid.
Fixed dimension	A single capacity fixed before the experiments.
FONDUE	Exponential-then-binary search for the largest capacity at which the estimated dimensions of the sampled and the mean representation differ by no more than a fifth of the data estimate; threshold computed once and held. Epoch budget per dataset follows the source, determined for the datasets it does not cover by the procedure the source describes.
FONDUE-VAR	Double capacity from twice the data estimate until unused variable types appear; return the count of the remaining ones. Run with both declared settings.
GECO + L_0	Gated latent coordinates, penalised by the expected number of open gates, under a reconstruction bound enforced by a Lagrange multiplier. <i>Deviation:</i> the source uses an augment-reinforce-merge gradient estimator; we use the hard-concrete relaxation of the same L_0 objective.
ARD-VAE	Hierarchical prior with per-axis variance fitted to the encoded data; relevant axes are those accounting for 99% of a sensitivity-weighted variance score. <i>Deviation:</i> the source uses the marginalised Student’s t target; we use a Gaussian with the same conjugate variance estimate.
ID-guided + target	Reference models, estimation, reliability checks, then every capacity in the window at or below twice the estimate, with a declared expansion rule; returns the smallest evaluated capacity meeting the quality target.
Ascending scan + target	The same objective without estimation or reference models: scan the grid upwards and stop at the first capacity meeting the target.

5.4. Provenance

Each run records the interpreter, framework, CUDA and cuDNN versions, the driver and device, the platform, a hash of the code and of the split indices, the objective convention, elapsed time, update count, termination state and convergence verdict. The earlier version of this work reported none of these for its training environment, and that environment can no longer be reconstructed; the numbers reported here come from the re-run.

6. Results

All results below use five independent training seeds per condition. The full procedure — reference models, estimation, selection and the final training at the selected dimension — is repeated for each seed; re-evaluating one checkpoint several times is not counted as several seeds. Unless stated otherwise the reported reconstruction error is the per-pixel mean squared error on the validation split, computed in evaluation mode from the deterministic reconstruction. The test split is used only once, on the checkpoint that validation already selected.

6.1. Method-Level Comparison

Table 2 reports, for each method, the selected dimension, the reconstruction error of the final model, the total wall-clock cost and the breakdown of termination states. Costs

include reference-model training, estimation, candidate training, the anchor, any re-search, probe fitting and evaluation; every method is charged up to the point where a selected model is ready for final evaluation. Trials that exhausted the budget are reported in the breakdown and are not dropped from the averages.

Three observations follow. First, guidance by an intrinsic-dimension estimate was the most expensive procedure we ran on MNIST, and it did not achieve the best reconstruction error. Second, adding the same quality criterion to a published selection algorithm produced a model that was both better and cheaper than the guided procedure. Third, an ascending scan over the candidate grid under that same criterion — using no intrinsic-dimension estimate and no reference models — reached the same selected dimension and the same error at a fraction of the cost, and succeeded on datasets where the guided procedure declined to apply. Figure 1 shows the quality–cost trade-off with all regularisation settings overlaid.

Table 2. Method-level comparison at the primary regularisation setting. Selected dimension is mean $\pm$ SD over five independent seeds; reconstruction error is the per-pixel mean on the validation split for the final model; cost is total wall-clock including reference models, estimation, candidates, the reference capacity, re-search and evaluation. Termination states are reported in full and no trial is dropped.

Method	Selected	Val. MSE	Cost (s)	Termination
<i>MNIST</i>				
Grid search (reconstruction)	12.8 $\pm$ 6.4	0.02098	628	success $\times$ 5
FONDUE + quality target	8.0 $\pm$ 0.0	0.02172	126	success $\times$ 5
FONDUE	7.0 $\pm$ 0.0	0.02193	41	success $\times$ 5
GECO + L_0	7.2 $\pm$ 0.4	0.02240	215	success $\times$ 5
ARD-VAE	6.0 $\pm$ 0.0	0.02244	176	success $\times$ 5
Ascending scan + target	6.0 $\pm$ 0.0	0.02244	182	success $\times$ 5
ID-guided + target	6.0 $\pm$ 0.0	0.02244	715	success $\times$ 5
<i>dSprites</i>				
Grid search (reconstruction)	57.6 $\pm$ 14.3	0.00116	2565	success $\times$ 5
Ascending scan + target	20.0 $\pm$ 4.0	0.00126	1797	success $\times$ 5
FONDUE + quality target	20.0 $\pm$ 4.0	0.00126	1676	success $\times$ 4, budget $\times$ 1
ARD-VAE	9.2 $\pm$ 1.5	0.00192	792	success $\times$ 5
FONDUE	4.2 $\pm$ 0.4	0.00550	141	success $\times$ 5
GECO + L_0	64.0 $\pm$ 0.0	0.00117	969	success $\times$ 5
ID-guided + target	64.0 $\pm$ 0.0	0.00117	1001	unreliable $\times$ 5
<i>CIFAR-10</i>				
Grid search (reconstruction)	140.8 $\pm$ 53.5	0.01383	428	success $\times$ 5
Ascending scan + target	24.0 $\pm$ 0.0	0.01512	157	success $\times$ 5
FONDUE + quality target	24.0 $\pm$ 0.0	0.01512	133	success $\times$ 5
GECO + L_0	20.0 $\pm$ 1.0	0.01629	400	success $\times$ 5
FONDUE	14.4 $\pm$ 0.5	0.01779	57	success $\times$ 5
ARD-VAE	14.2 $\pm$ 0.4	0.01709	314	success $\times$ 5
ID-guided + target	131.2 $\pm$ 113.9	0.01401	499	unreliable $\times$ 5

On MNIST the guided procedure cost 715 s, more than any other method we ran and 17 times the 41 s of the selection algorithm we compare against, while reaching a slightly worse reconstruction error (0.02244 against 0.02193). Adding the quality target to that algorithm gave both a better error and a lower cost than the guided procedure (0.02172 at 126 s). The ascending scan returned exactly the same dimension and the same error as the guided procedure at 182 s, a quarter of its cost, using no intrinsic-dimension estimate and no reference models.

On the other two datasets the guided procedure did not apply at all: its estimation step failed the reliability check on every seed and it fell back to a conventional search, which

on CIFAR-10 was itself unstable (131.2 ± 113.9). The ascending scan succeeded on both, selecting 20.0 ± 4.0 on dSprites and 24.0 ± 0.0 on CIFAR-10.

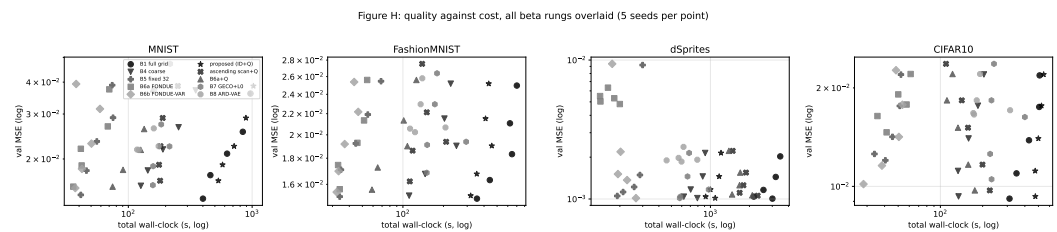


Figure 1. Reconstruction error against total cost, with all five regularisation settings overlaid. Five seeds per point; both axes logarithmic.

6.2. Sensitivity of the Intrinsic-Dimension Estimate

The quantity the guided procedure depends on is not a property of the data alone. Estimating it in the input space, in a PCA-preprocessed space and through trained reference autoencoders gives values that differ by a factor of between about 1.7 and 3.7 on the same dataset (Figure 2). Because the candidate window is placed at twice the estimate, the window moves by the same factor.

The agreement between two standard estimators also depends on the size of the estimation sample. With a thousand points all four datasets pass the agreement criterion we fixed in advance; with ten thousand points only one does. The decision to apply or to decline the procedure therefore depends on a quantity that is free to choose. We fixed the sample size before seeing any result, and we report the dependence rather than selecting the value that would have let the procedure proceed.

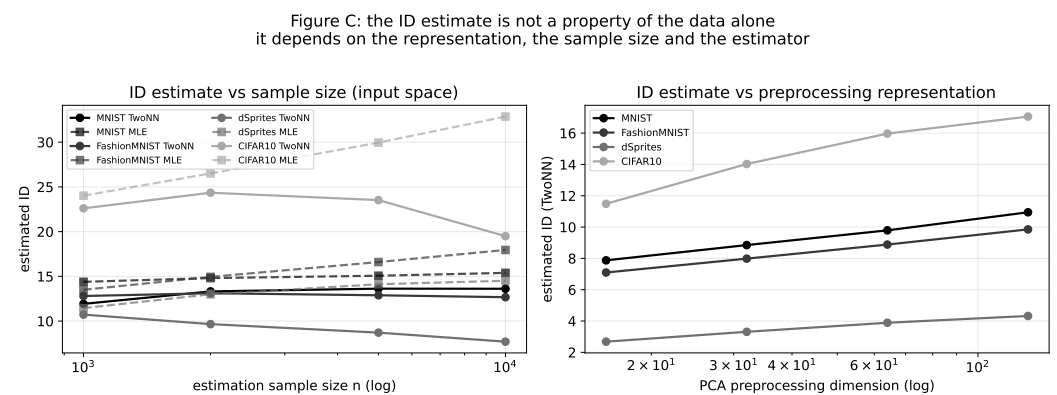


Figure 2. The intrinsic-dimension estimate as a function of the estimation sample size (left) and of the preprocessing representation (right).

On dSprites the estimate is 7.70 in the input space, between 2.68 and 4.32 in principal-component spaces of increasing rank, and between 2.08 and 4.56 through trained reference autoencoders of three capacities: a range of 3.7 to one on a single dataset. On MNIST the corresponding range is 1.7 to one. The relative disagreement between the two estimators rises from 0.068 at a thousand points to 0.885 at ten thousand on dSprites, and from 0.063 to 0.686 on CIFAR-10.

6.3. Threshold Governance and Its Ordering

Varying the regularisation weight over five settings moves the dimension that each method selects. Table 3 reports the span of the selected dimension across that ladder, and Figure 3 plots it. The spans differ by two orders of magnitude between methods, and the ordering is consistent across datasets.

One method is flat. The automatic-relevance variant returns an identical value at every rung on every dataset, because it has no free threshold to vary: the relevance of each axis is read from a hierarchical prior fitted to the data. Every other method we examined — including plain grid search under a reconstruction criterion — moves substantially. This is the clearest single result of the study: the question is not whether a selection rule has a free parameter, but how strongly the answer depends on it.

Table 3. Span of the selected dimension across five regularisation settings, by method and dataset. Smaller is less governed by the free parameter; five seeds per setting.

Method	MNIST	Fashion	dSprites	CIFAR-10	Mean
Fixed dimension	0.0	0.0	0.0	0.0	0.0
ARD-VAE	0.0	0.0	0.0	0.0	0.0
FONDUE	3.8	0.8	0.8	8.8	3.5
GECO + L_0	8.8	2.0	0.0	34.2	11.2
Ascending scan + target	6.0	3.6	15.2	52.8	19.4
FONDUE + quality target	3.6	1.6	28.0	44.8	19.5
Grid search	14.8	20.8	40.0	91.2	41.7
ID-guided + target	6.0	37.2	0.0	214.4	64.4
Coarse search	33.6	37.2	0.0	214.4	71.3
FONDUE-VAR	50.6	63.0	106.2	68.8	72.2

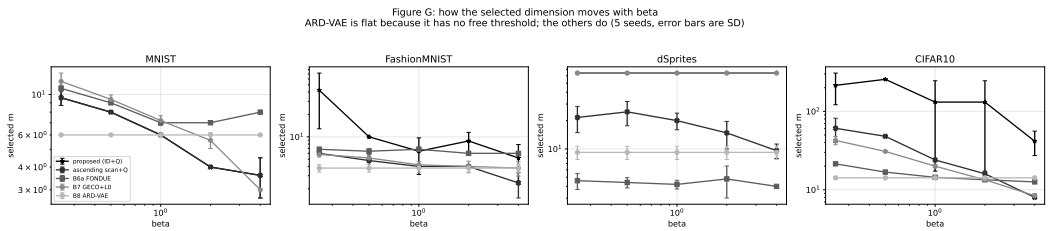


Figure 3. Selected dimension against the regularisation weight, five seeds, error bars are standard deviations. The automatic-relevance method is flat on every dataset.

The spans differ by more than two orders of magnitude, from zero to 72.2 on average. Grid search under a reconstruction criterion, the least elaborate rule in the comparison, has a mean span of 41.7: the premise that fixing the regularisation weight fixes an optimal dimension does not hold even there. The guided procedure sits at 64.4, inflated by CIFAR-10, where it inherits the instability of the search it falls back to.

6.4. Stability of the Activity Readout

We compared two readouts computed from the same checkpoints: the number of latent coordinates whose posterior means retain variance, and the variable-type classification used by a published algorithm. Both are unstable early in training and both stabilise by the end; by the final checkpoint they agree closely, so the claim that one is more stable than the other is not supported. At the earliest checkpoints the ordering is in fact reversed on one condition.

The activity readout is governed by its own threshold. On natural images the verdict reverses within the range of threshold values in common use: at the small end no truncation is detected at any capacity, and at the large end the count saturates well below capacity (Figure 4). Normalising the threshold by the scale of the representation does not remove this dependence; it reparameterises it.

We also asked whether the activity readout carries information that other diagnostics do not, using a prediction task on held-out conditions defined in advance. The pre-

registered rule returned a positive verdict. A control that adds the same number of random features to the same baseline produced an improvement that cannot be distinguished from it. We therefore report that the pre-registered test lacked the power to attribute the improvement to the readout, and we do not claim incremental value.

Figure D: the AU verdict is governed by the threshold delta on CIFAR-10 the conclusion reverses between delta=0.001 and delta=0.1 (beta=1, 5 seeds)

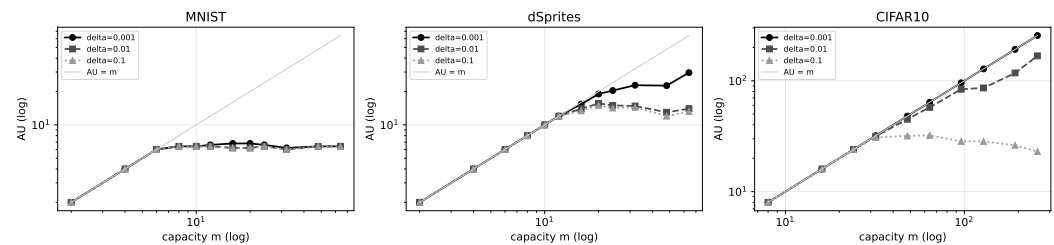


Figure 4. Activity count against capacity for three settings of the activity threshold. On CIFAR-10 the verdict reverses between the smallest and largest setting.

On CIFAR-10 at the smallest threshold the count equals capacity in every condition we ran, so no truncation is detected anywhere; at the largest it saturates between 23 and 32 while capacity grows to 256. The pre-registered prediction task gave a mean improvement in area under the curve of $+0.108$ with a bootstrap interval of $[+0.059, +0.157]$, which the registered rule reads as incremental value; a control adding three random features to the same baseline gave $+0.098$ with interval $[-0.025, +0.227]$, which the first interval does not exclude.

6.5. Natural Images

On the two datasets other than MNIST for which the guided procedure was run, the estimation step failed its own reliability checks on every seed and the procedure declined to apply, falling back to the pre-declared conventional search. The fallback was itself unstable on natural images. The procedure succeeded only on the dataset used for development. We therefore do not claim applicability to natural images.

On dSprites the three reference models gave estimates of 3.96, 3.91 and 2.10, a relative range of 0.474 against a criterion of 0.15, and the two estimators disagreed by 1.816 against a criterion of 0.25. On CIFAR-10 the checks failed likewise. In both cases the procedure returned the unreliable state on all five seeds.

6.6. Implementation Failure Modes

Three defects surfaced during this study, all of the same kind: a sum was written as a mean, or the reverse, so that two terms of an objective were weighted relative to each other by a factor of the order of the dimensionality. None of them caused training to fail; each produced plausible curves and would not have been visible in the reported numbers.

The first was in our own objective and affected every variational experiment in the earlier version of this work. The second appeared in our re-implementation of a pruning method and caused it to violate its own reconstruction constraint while reporting a selected dimension. The third is a degenerate output of a published algorithm under one of its declared settings, which we reproduce faithfully and report as such rather than as an error of that work. We describe all three, with the checks that now guard against them, because this class of defect is invisible in aggregate results and is a realistic hazard for anyone re-implementing these methods.

In the first case the divergence term was averaged over latent dimensions as well as over the batch while the reconstruction term was summed over input dimensions, so the effective weight was the nominal weight divided by the capacity. The previously reported

dependence of the activity count on capacity is reproduced to within 1.7 units by re-reading it as a function of that ratio. In the second, a reconstruction constraint was evaluated as a per-pixel mean where its source defines it as a sum over pixels, leaving it smaller by the input dimension than the term it was meant to balance; the method then pruned from 64 dimensions to 1 while violating the constraint it reports satisfying. In the third, the variable-type algorithm under one of its two declared settings returns an active count of zero on MNIST at the epoch budget its authors use, on all five seeds.

7. Discussion

7.1. *What the Comparison Shows*

The question we set out to answer was which of the available signals supports the choice of a latent dimension. On the evidence collected here, the intrinsic-dimension estimate does not. It is expensive to obtain, it depends on the representation in which it is computed, and a search that ignores it entirely reaches the same answer sooner. The activity readout does not either: it agrees with the alternative readout once training has finished, its verdict on natural images reverses with its own threshold, and in a prediction task designed in advance it was not distinguishable from random features of the same number.

What remains is less a method than a measurement. Almost every rule we examined is governed by a free parameter, and the dimension it selects moves severalfold as that parameter varies. The spans in Table 3 differ by two orders of magnitude between methods, and the ordering reproduces across datasets. That ordering is the most transferable thing we found.

7.2. *Relation to Existing Work*

Two points of overlap deserve to be stated plainly rather than discovered by a reader. First, the appendix of the intrinsic-dimension algorithm we compare against already proposes selecting the dimension from the variable types of a briefly trained model, which is the same idea as the activity-based verification we set out to study; its authors set it aside for reasons our measurements reproduce. Second, the pruning method we added during revision already selects the smallest capacity that satisfies an explicit reconstruction constraint. The quality criterion we use throughout this paper is the same construction. We therefore do not claim it as a contribution, and we report the correspondence quantitatively: the dimension that method selects tracks the constraint we set for it across the whole ladder.

7.3. *What Would Change the Conclusion*

These are observations under one architecture family per dataset, one candidate grid, five seeds and five regularisation settings. They do not establish that intrinsic dimension is uninformative in general; they establish that under these conditions it bought nothing that a cheaper search did not. A different quality target, a capacity range where reconstruction still improves steeply, or a dataset whose estimate is stable across representations could each change the balance. The measurement we would trust to transfer is the sensitivity ordering, not the individual selected dimensions.

Two comparisons remain absent. Our re-implementation of the intrinsic-dimension algorithm does not reproduce its published value on the one dataset where a published value exists, so that value stays in a reference column and the gap stands. The pruning method is implemented with a different gradient estimator than its authors used, and the relevance method with a Gaussian approximation in place of the marginalised prior; both deviations are stated where the methods are defined.

7.4. Synthetic Versus Real Data

On standardised synthetic manifolds with a known embedding dimension the activity readout behaves as the topological argument would suggest: it saturates at the embedding dimension, and zeroing the coordinates it excludes changes reconstruction by less than a fifth of a percent. An earlier version of this work reported the opposite on one manifold; that observation was an artefact of leaving the coordinates unstandardised, where one axis carried two parts in a thousand of the variance and could be ignored at almost no cost. The contrast between the synthetic case, where the readout works, and the image datasets, where its verdict is set by its threshold, is itself the finding we would carry forward.

8. Conclusions

We compared the signals that have been proposed for choosing the latent dimension of a variational autoencoder, under a protocol fixed before the experiments and across four datasets, five seeds and five regularisation settings. Guidance by an intrinsic-dimension estimate conferred no measurable advantage over an ascending search under the same quality criterion, and declined to apply on the image datasets it was not developed on. The activity-based readout was not more stable than the alternative it was meant to improve on, and carried no information beyond a matched set of random features. The quality criterion that did transfer is not ours; it coincides with the constraint used by an existing pruning method.

What we can offer is a measurement rather than a method: a reproducible ordering of how strongly each rule's answer depends on its free parameter, with one method at zero and another spanning more than a hundred dimensions on the same data. We also record three implementation defects of a single kind, invisible in aggregate results, that anyone re-implementing these methods is likely to meet.

1. Kingma, D.P.; Welling, M. Auto-encoding variational Bayes. In *Proceedings of the 2nd International Conference on Learning Representations (ICLR)*, Banff, AB, Canada, 14–16 April 2014. arXiv:1312.6114.
2. Higgins, I.; Matthey, L.; Pal, A.; Burgess, C.; Glorot, X.; Botvinick, M.; Mohamed, S.; Lerchner, A. β -VAE: Learning basic visual concepts with a constrained variational framework. In *Proceedings of the 5th International Conference on Learning Representations (ICLR)*, Toulon, France, 24–26 April 2017.
3. Bengio, Y.; Courville, A.; Vincent, P. Representation learning: A review and new perspectives. *IEEE Trans. Pattern Anal. Mach. Intell.* **2013**, *35*, 1798–1828. <https://doi.org/10.1109/TPAMI.2013.50>.
4. Fefferman, C.; Mitter, S.; Narayanan, H. Testing the manifold hypothesis. *J. Am. Math. Soc.* **2016**, *29*, 983–1049. <https://doi.org/10.1090/jams/852>.
5. Tenenbaum, J.B.; de Silva, V.; Langford, J.C. A global geometric framework for nonlinear dimensionality reduction. *Science* **2000**, *290*, 2319–2323. <https://doi.org/10.1126/science.290.5500.2319>.
6. Roweis, S.T.; Saul, L.K. Nonlinear dimensionality reduction by locally linear embedding. *Science* **2000**, *290*, 2323–2326. <https://doi.org/10.1126/science.290.5500.2323>.
7. Tipping, M.E.; Bishop, C.M. Probabilistic principal component analysis. *J. R. Stat. Soc. Ser. B* **1999**, *61*, 611–622. <https://doi.org/10.1111/1467-9868.00196>.
8. Bishop, C.M. Bayesian PCA. In *Advances in Neural Information Processing Systems*, Vol. 11, 1999; pp. 382–388.
9. Facco, E.; d'Errico, M.; Rodriguez, A.; Laio, A. Estimating the intrinsic dimension of datasets by a minimal neighborhood information. *Sci. Rep.* **2017**, *7*, 12140. <https://doi.org/10.1038/s41598-017-11873-y>.
10. Levina, E.; Bickel, P.J. Maximum likelihood estimation of intrinsic dimension. In *Advances in Neural Information Processing Systems (NeurIPS)*, Vol. 17, 2005.

11. Ceruti, C.; Bassis, S.; Rozza, A.; Lombardi, G.; Casiraghi, E.; Campadelli, P. DANCo: An intrinsic dimensionality estimator exploiting angle and norm concentration. *Pattern Recognit.* **2014**, *47*, 2569–2581. <https://doi.org/10.1016/j.patcog.2014.02.013>.
12. Johnsson, K.; Soneson, C.; Fontes, M. Low bias local intrinsic dimension estimation from expected simplex skewness. *IEEE Trans. Pattern Anal. Mach. Intell.* **2015**, *37*, 196–202. <https://doi.org/10.1109/TPAMI.2014.2343220>.
13. Camastra, F.; Staiano, A. Intrinsic dimension estimation: Advances and open problems. *Inf. Sci.* **2016**, *328*, 26–41. <https://doi.org/10.1016/j.ins.2015.08.029>.
14. Allegra, M.; Facco, E.; Denti, F.; Laio, A.; Mira, A. Data segmentation based on the local intrinsic dimension. *Sci. Rep.* **2020**, *10*, 16449. <https://doi.org/10.1038/s41598-020-72222-0>.
15. Costa, J.A.; Hero, A.O. Geodesic entropic graphs for dimension and entropy estimation in manifold learning. *IEEE Trans. Signal Process.* **2004**, *52*, 2210–2221. <https://doi.org/10.1109/TSP.2004.831130>.
16. Ansuini, A.; Laio, A.; Macke, J.H.; Zoccolan, D. Intrinsic dimension of data representations in deep neural networks. In *NeurIPS 2019*, Vol. 32, 2019.
17. Pope, P.; Zhu, C.; Abdelkader, A.; Goldblum, M.; Goldstein, T. The intrinsic dimension of images and its impact on learning. In *ICLR 2021*, 2021.
18. Bonheme, L.; Grzes, M. FONDUE: An algorithm to find the optimal dimensionality of the latent representations of variational autoencoders. *arXiv* **2022**, arXiv:2209.12806. <https://doi.org/10.48550/arXiv.2209.12806>.
19. Bonheme, L.; Grzes, M. Be more active! Understanding the differences between mean and sampled representations of variational autoencoders. *J. Mach. Learn. Res.* **2023**, *24*, 1–30.
20. De Boom, C.; Wauthier, S.; Verbelen, T.; Dhoedt, B. Dynamic narrowing of VAE bottlenecks using GECO and L_0 regularization. *arXiv* **2020**, arXiv:2003.10901. <https://doi.org/10.48550/arXiv.2003.10901>.
21. Rezende, D.J.; Viola, F. Taming VAEs. *arXiv* **2018**, arXiv:1810.00597. <https://doi.org/10.48550/arXiv.1810.00597>.
22. Louizos, C.; Welling, M.; Kingma, D.P. Learning sparse neural networks through L_0 regularization. In *Proceedings of the 6th International Conference on Learning Representations (ICLR)*, Vancouver, BC, Canada, 30 April–3 May 2018. arXiv:1712.01312.
23. Saha, S.; Joshi, S.; Whitaker, R. ARD-VAE: A statistical formulation to find the relevant latent dimensions of variational autoencoders. *arXiv* **2025**, arXiv:2501.10901. <https://doi.org/10.48550/arXiv.2501.10901>.
24. Lucas, J.; Tucker, G.; Grosse, R.B.; Norouzi, M. Don't blame the ELBO! A linear VAE perspective on posterior collapse. In *NeurIPS 2019*, Vol. 32, 2019.
25. Burda, Y.; Grosse, R.; Salakhutdinov, R. Importance weighted autoencoders. In *Proceedings of the 4th International Conference on Learning Representations (ICLR)*, San Juan, Puerto Rico, 2–4 May 2016. arXiv:1509.00519.
26. Hoffman, M.D.; Johnson, M.J. ELBO surgery: Yet another way to carve up the variational evidence lower bound. In *NIPS Workshop on Advances in Approximate Bayesian Inference*, Barcelona, Spain, 2016.
27. Fu, H.; Tan, C.; Peng, B.; Zhao, Z.; Yan, T.; Chang, C.; Zhao, L. Cyclical annealing schedule: A simple approach to mitigating KL vanishing. In *Proceedings of NAACL-HLT 2019*, Minneapolis, MN, USA, 2019; pp. 240–250.
28. Rolinek, M.; Zietlow, D.; Martius, G. Variational autoencoders pursue PCA directions (by accident). In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, Long Beach, CA, USA, 15–20 June 2019; pp. 12406–12415. <https://doi.org/10.1109/CVPR.2019.01269>.
29. Hatcher, A. *Algebraic Topology*; Cambridge University Press: Cambridge, UK, 2002.
30. Cover, T.M.; Thomas, J.A. *Elements of Information Theory*, 2nd ed.; Wiley-Interscience: Hoboken, NJ, USA, 2006.
31. LeCun, Y.; Bottou, L.; Bengio, Y.; Haffner, P. Gradient-based learning applied to document recognition. *Proc. IEEE* **1998**, *86*, 2278–2324. <https://doi.org/10.1109/5.726791>.
32. Xiao, H.; Rasul, K.; Vollgraf, R. Fashion-MNIST: A Novel Image Dataset for Benchmarking Machine Learning Algorithms. *arXiv* **2017**, 1708.07747.

33. Matthey, L.; Higgins, I.; Hassabis, D.; Lerchner, A. dSprites: Disentanglement testing sprites dataset. 2017. Available online: <https://github.com/google-deepmind/dsprites-dataset> (accessed on 12 September 2026). 473 474 475
34. Krizhevsky, A. Learning multiple layers of features from tiny images. *Technical Report*; University of Toronto: Toronto, ON, Canada, 2009. 476 477
35. Obata, C.; Jin'no, K. Analysis of the relationship between manifold structure and latent space capacity of input data using VAE. In *Proceedings of the 2025 International Symposium on Nonlinear Theory and Its Applications (NOLTA2025)*, Okinawa, Japan, 27–31 October 2025; pp. 32–35. 478 479 480